Research Article

Abstract

Place abstract text here

Keywords: Place keywords here

1. INTRODUCTION

Place text here

2. LITERATURE REVIEW

[1] For semantic space modelling.

3. METHOD AND MODEL

3.1. Phase 0: Problem Definition

3.1.1. Define the drift problem in one paragraph

This project visualises concept drift in a HyperRAG system operating over simple medical datasets. As the system answers a query using a HyperRAG network, a sparse autoencoder identifies the concepts it relies on, and complex (quantum-inspired) linear algebra gives each concept a magnitude and an angle in Hilbert space; magnitude for how strongly the concept is present, angle for its relation to other concepts. Plotting these across queries turns the similarities and differences between concepts into visible geometry: as the query changes, the relative angles between concepts open and close, and that movement is the drift. The aim is to make the system's shifting internal concepts legible in a way a real-valued, magnitude-only view cannot. We want to see how the system judges relations in a Hilbert space, and how those relations change over different prompts. We also investigate whether representing concepts with complex-valued (quantum-inspired) linear algebra can reveal information about polysemy and concept relationships that a real-valued vector space cannot.[1].

3.1.2. Identify target audience

Researchers, engineers and artists.

3.1.3. Define expected output (visualiser + baseline comparison)

A 3D render (a projection of the high-dimensional Hilbert space) of concepts moving as queries change, alongside a baseline of the same concepts in real vector space for comparison.

3.1.4. Write success criteria

The Hilbert-space view (angle + magnitude) reveals relational structure or drift between concepts that the real-valued, magnitude-only baseline does not. Equivalently; degrading the Hilbert-space representation down to a real vector space visibly loses information, and the visualiser makes that lost information apparent.

3.1.5. Record assumptions / risks

Information loss and spurious concepts in both the real and Hilbert-space representations, and the risk of overfitting to a small dataset.

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3.2. Phase 1: Hypergraph RAG

3.2.1. Hypergraph Store

This section takes our initial data and stores it as a hypergraph, each entity, is a node and n-ary of facts are hyperedges. This data strcuture is built as a standalone component, hyperNetX will be added downstream. This section has no quantum linear algebra, this is the store of our preliminary data structure.

3.2.2. Hypergraph persistence

Responsible for writing and reading the hypergraph to disk. This is done using JSON, and the hypergraph is stored as a dictionary of nodes and edges.

3.2.3. Hypergraph Schema

Defines the structure and types of nodes and edges in the hypergraph. Defines what is an entity and what is a relaitonship, and how they are connected. This is important when the RAG system wants to query the network.

Data Representation We model the knowledge base as a hypergraph $H = (V, E)$. Entities are nodes and relationships are *role-typed* hyperedges.

Entities. Let V = set of entities. Each $v \in V$ carries a stable identifier and optional attributes; the identifier is the anchor to which extracted concepts are later bound.

Relationships. Let $\mathcal{R}$ be a set of relation labels and Φ a set of role labels. A hyperedge is a triple

$$e = (\ell,\ r,\ \rho), \qquad \ell \in \mathcal{L}, \quad r \in \mathcal{R}, \quad \rho\colon \Phi \rightharpoonup V, \quad |\operatorname{dom}(\rho)| \geq 2,$$

where

- ℓ is a unique identifier drawn from a label set $\mathcal{L}$ (the stable anchor for the fact);
- $r \in \mathcal{R}$ is the relation label naming the relationship (e.g. TREATS);
- $\rho\colon \Phi \rightharpoonup V$ is a partial map from role labels to entities, assigning each participating entity the role it plays;
- $\operatorname{dom}(\rho) \subseteq \Phi$ is the set of roles actually used by this edge, and $|\operatorname{dom}(\rho)| \geq 2$ requires at least two role-bound entities.

Eg. "aspirin treats headache at a 300 mg dose" is a single hyperedge

$$e_1 = \big(\texttt{he\_0001},\ \textsc{treats},\ \{\,\mathsf{agent} \mapsto \text{aspirin},\ \mathsf{target} \mapsto \text{headache},\ \mathsf{dose} \mapsto 300\,\text{mg}\,\}\big).$$

The role map preserves directionality: $\rho(\mathsf{agent}) \neq \rho(\mathsf{target})$ distinguishes "X treats Y" from "Y treats X", and $|\operatorname{dom}(\rho)| \geq 2$ lets one hyperedge hold an n-ary fact rather than $\binom{n}{2}$ binary edges.

The schema *declares* and validates a supplied edge:

$$\operatorname{valid}(e) \iff r \in \mathcal{R} \ \wedge\ |\operatorname{dom}(\rho)| \geq 2.$$

Identifying edges from source data $\mathcal{D}$ is a separate upstream map $\mathcal{E}\colon \mathcal{D} \to 2^E$, disjoint from this component.

3.3. Phase 2: Complex Embedding Layer

Where entities and relations become embeddings, embed these in a Hilbert space. We make use of RotatE encodings, to apply role-driven rotations to each concept.

3.3.1. RotatE encoding (discrete, role-driven)

$$R_\rho = \mathrm{diag}\Big(e^{i\theta_1^\rho}, \ldots, e^{i\theta_d^\rho}\Big), \qquad c = \frac{1}{k}\sum_{i=1}^{k} R_{\rho_i}\, e_i. \tag{1}$$

- $e_i \in \mathbb{C}^d$ — complex embedding of the i-th entity (d “phasors”)
- $\theta^\rho \in \mathbb{R}^d$ — the phase angles of role ρ (one per dimension)
- R_ρ — a diagonal *unitary*: it rotates each phasor of an entity by its role angle
- k — the number of role-bound entities in the fact
- $c \in \mathbb{C}^d$ — the encoded concept — the aggregate placement

We make use of a diagonal matrix, the rotation is applied to each dimension of the embedding independently. We rotate the phase of the specific dimension by its own angle θ_j^ρ. We have neglected to add positional data to the hypergraph, hence using the diagonal matrix to rotate each dimension independently. In future work we can explore giving the concept walks eigenvalues a RoPE-like frequency spectrum to explore how fast modes shift.

3.3.2. Encoding as a quantum state

$$|\psi(0)\rangle = \frac{c}{\|c\|}, \qquad \langle\psi|\psi\rangle = 1, \qquad p_k = |\psi_k|^2. \tag{2}$$

- $|\psi(0)\rangle$ — the unit-norm state — the initial condition of the walk
- $\|c\|$ — the Euclidean length of c (what we divide by to normalise)
- $\langle\psi|\psi\rangle$ — the state’s inner product with itself; $= 1$ means unit length
- $p_k = |\psi_k|^2$ — the Born-rule probability of component k

Normalising has turned this concept vector into a genuine wave function, The unit normal state is the wave function of the concept at time T=0,

The born-rule probability is the probability of measuring a specific concept, this is the magnitude of the complex number squared.

This gives concepts encoded in Hilbert space a probabilistic interpretation, and allows us to use interpretations from quantum mechanics to understand the behaviour of concepts in the system.

This has set where our clock hands start at time T=0, and we can now evolve the concept over time using a unitary operator.

The multiplaction of the Hamiltonian by the imaginary unit i and the negative sign in the exponent ensures that the evolution is unitary, preserving the probability distribution of the state over time.

3.4. Phase 3: The concept walk, Evolution of concepts over time

3.4.1. Unitary evolution (continuous, time-driven)

$$i\,\frac{d}{dt}|\psi(t)\rangle = H\,|\psi(t)\rangle \qquad \Longrightarrow \qquad |\psi(t)\rangle = e^{-iHt}\,|\psi(0)\rangle. \tag{3}$$

$$H = H^\dagger \quad \Longrightarrow \quad \left(e^{-iHt}\right)^\dagger e^{-iHt} = I \quad \Longrightarrow \quad \|\psi(t)\| = 1 \quad \text{for all } t. \tag{4}$$

- H — the Hermitian Hamiltonian ($d \times d$) — the *generator* of the evolution
- $H^\dagger$ — the conjugate transpose of H; "$H = H^\dagger$" means H is Hermitian
- e^{-iHt} — the *unitary propagator* — "the clock" that advances the state
- t — continuous time
- I — the identity matrix

Now the clock runs. Because H is Hermitian, the propagator e^{-iHt} is *unitary*, so the norm is preserved for *all* time: the state rotates but never shrinks. This is the non-degradation guarantee — and it is exactly what "unitary" means.

4. RESULTS AND DISCUSSION

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5. CONCLUSION AND FUTURE WORK

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6. DECLARATION

Conflict of Interest: Place text here

Author Contribution: Place text here

7. REFERENCES

[1] Timo Aukusti Laine. "The Quantum LLM: Modeling Semantic Spaces with Quantum Principles". In: *OA Journal of Applied Science and Technology* 3.2 (2025), pp. 1–13. DOI: 10.48550/arXiv.2504.13202. arXiv: 2504.13202 [cs.AI]. URL: https://doi.org/10.48550/arXiv.2504.13202.